**Table 2-3** **Magnitude and phase of the reflection coefficient for various types of loads. The normalized load impedance $z_L = Z_L/Z_0 = (R + jX)/Z_0 = r + jx$, where $r = R/Z_0$ and $x = X/Z_0$ are the real and imaginary parts of $z_L$, respectively.**

**Reflection Coefficient $\Gamma = |\Gamma| e^{j\theta_r}$**

| Load | $\lvert\Gamma\rvert$ | $\theta_r$ |
|---|---|---|
| $Z_0$ — $Z_L = (r + jx)Z_0$ | $\left[\frac{(r-1)^2 + x^2}{(r+1)^2 + x^2}\right]^{1/2}$ | $\tan^{-1}\left(\frac{x}{r-1}\right) - \tan^{-1}\left(\frac{x}{r+1}\right)$ |
| $Z_0$ — $Z_0$ | 0 (no reflection) | irrelevant |
| $Z_0$ — (short) | 1 | $\pm 180°$ (phase opposition) |
| $Z_0$ — (open) | 1 | 0 (in-phase) |
| $Z_0$ — $jX = j\omega L$ | 1 | $\pm 180° - 2\tan^{-1} x$ |
| $Z_0$ — $jX = \frac{-j}{\omega C}$ | 1 | $\pm 180° - 2\tan^{-1} x$ |

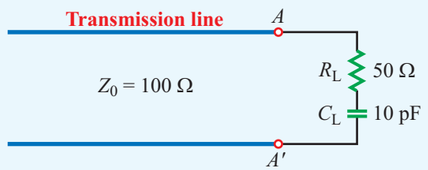



**Figure 2-13** RC load (Example 2-3).

The normalized load impedance is

$$\begin{aligned} z_L &= \frac{Z_L}{Z_0} \\ &= \frac{R_L - j/(\omega C_L)}{Z_0} \\ &= \frac{1}{100}\left(50 - j\frac{1}{2\pi \times 10^8 \times 10^{-11}}\right) \\ &= (0.5 - j1.59)\ \Omega. \end{aligned}$$

From Eq. (2.59), the voltage reflection coefficient is

$$\begin{aligned} \Gamma &= \frac{z_L - 1}{z_L + 1} \\ &= \frac{0.5 - j1.59 - 1}{0.5 - j1.59 + 1} \\ &= \frac{-0.5 - j1.59}{1.5 - j1.59} \\ &= \frac{-1.67 e^{j72.6°}}{2.19 e^{-j46.7°}} \\ &= -0.76 e^{j119.3°}. \end{aligned}$$

This result may be converted into the form of Eq. (2.62) by replacing the minus sign with $e^{-j180°}$. Thus,

$$\begin{aligned} \Gamma &= 0.76 e^{j119.3°} e^{-j180°} \\ &= 0.76 e^{-j60.7°} \\ &= 0.76 \angle -60.7°, \end{aligned}$$

or

$$|\Gamma| = 0.76, \qquad \theta_r = -60.7°.$$





表 2-3 各种负载的反射系数的大小和相位。归一化负载阻抗 CE
zL = $Z_L/Z_0$ = (R + jX)/$Z_0$ = r + jx，其中 r = R/$Z_0$ 和 x = X/$Z_0$ 分别是 $z_L$ 的实部和虚部。

**Reflection Coefficient $\Gamma = |\Gamma| e^{j\theta_r}$**

| Load | $\lvert\Gamma\rvert$ | $\theta_r$ |
|---|---|---|
| $Z_0$ — $Z_L = (r + jx)Z_0$ | $\left[\frac{(r-1)^2 + x^2}{(r+1)^2 + x^2}\right]^{1/2}$ | $\tan^{-1}\left(\frac{x}{r-1}\right) - \tan^{-1}\left(\frac{x}{r+1}\right)$ |
| $Z_0$ — $Z_0$ | 0 (no reflection) | irrelevant |
| $Z_0$ — (short) | 1 | $\pm 180°$ (phase opposition) |
| $Z_0$ — (open) | 1 | 0 (in-phase) |
| $Z_0$ — $jX = j\omega L$ | 1 | $\pm 180° - 2\tan^{-1} x$ |
| $Z_0$ — $jX = \frac{-j}{\omega C}$ | 1 | $\pm 180° - 2\tan^{-1} x$ |

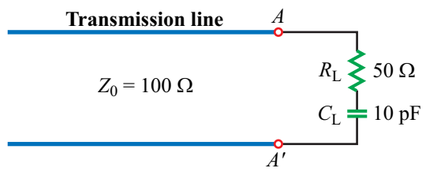



图 2-13 RC 负载（示例 2-3）。

The normalized load impedance is

$$\begin{aligned} z_L &= \frac{Z_L}{Z_0} \\ &= \frac{R_L - j/(\omega C_L)}{Z_0} \\ &= \frac{1}{100}\left(50 - j\frac{1}{2\pi \times 10^8 \times 10^{-11}}\right) \\ &= (0.5 - j1.59)\ \Omega. \end{aligned}$$

从方程。 (2.59)，电压反射系数为

$$\begin{aligned} \Gamma &= \frac{z_L - 1}{z_L + 1} \\ &= \frac{0.5 - j1.59 - 1}{0.5 - j1.59 + 1} \\ &= \frac{-0.5 - j1.59}{1.5 - j1.59} \\ &= \frac{-1.67 e^{j72.6°}}{2.19 e^{-j46.7°}} \\ &= -0.76 e^{j119.3°}. \end{aligned}$$

该结果可以转化为式（1）的形式。 (2.62) 用 $e^{-j180°}$ 替换减号。因此，

$$\begin{aligned} \Gamma &= 0.76 e^{j119.3°} e^{-j180°} \\ &= 0.76 e^{-j60.7°} \\ &= 0.76 \angle -60.7°, \end{aligned}$$

或者

$$|\Gamma| = 0.76, \qquad \theta_r = -60.7°.$$

**Example 2-4: $|\Gamma|$ for Purely Reactive Load**

Show that $|\Gamma| = 1$ for a lossless line connected to a purely reactive load.

**Solution:** The load impedance of a purely reactive load is $Z_L = jX_L$. From Eq. (2.59), the reflection coefficient is

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{jX_L - Z_0}{jX_L + Z_0} = \frac{-(Z_0 - jX_L)}{(Z_0 + jX_L)} = \frac{-\sqrt{Z_0^2 + X_L^2}\, e^{-j\theta}}{\sqrt{Z_0^2 + X_L^2}\, e^{j\theta}} = -e^{-j2\theta} = 1e^{j(\pi - 2\theta)},$$

where $\theta = \tan^{-1} X_L/Z_0$. Hence, for $\Gamma = |\Gamma|e^{j\theta_r}$,

$$|\Gamma| = 1$$

and

$$\theta_r = \pi - 2\theta = \pi - 2\tan^{-1}\left(\frac{X_L}{Z_0}\right).$$

**Exercise 2-8:** A 50 Ω lossless transmission line is terminated in a load with impedance $Z_L = (30 - j200)$ Ω. Calculate the voltage reflection coefficient at the load.

**Answer:** $\Gamma = 0.93/\underline{-27.5^\circ}$. (See EM.)

**Exercise 2-9:** A 150 Ω lossless line is terminated in a capacitor with impedance $Z_L = -j30$ Ω. Calculate Γ.

**Answer:** $\Gamma = 1/\underline{-157.4^\circ}$. (See EM.)

**Exercise 2-10:** Given that the reflection coefficient at the load is $\Gamma = 0.6 - j0.3$, determine the normalized load impedance $z_L$.

**Answer:** $z_L = 2.2 - j2.4$. (See EM.)

## 2-6.2 Standing Waves

Using the relation $V_0^- = \Gamma V_0^+$ in Eqs. (2.54a) and (2.54b) yields

$$\widetilde{V}(z) = V_0^+(e^{-j\beta z} + \Gamma e^{j\beta z}), \tag{2.63a}$$

$$\widetilde{I}(z) = \frac{V_0^+}{Z_0}(e^{-j\beta z} - \Gamma e^{j\beta z}). \tag{2.63b}$$

These expressions now contain only one (yet to be determined) unknown, $V_0^+$. Before we proceed to solve for $V_0^+$, however, let us examine the physical meaning underlying these expressions. We begin by deriving an expression for $|\widetilde{V}(z)|$, which is the magnitude of $\widetilde{V}(z)$. Upon using Eq. (2.62) in Eq. (2.63a) and applying the relation $|\widetilde{V}(z)| = [\widetilde{V}(z)\widetilde{V}^*(z)]^{1/2}$, where $\widetilde{V}^*(z)$ is the complex conjugate of $\widetilde{V}(z)$, we have

$$\begin{aligned}|\widetilde{V}(z)| &= \left\{\left[V_0^+(e^{-j\beta z} + |\Gamma|e^{j\theta_r}e^{j\beta z})\right] \cdot \left[(V_0^+)^*(e^{j\beta z} + |\Gamma|e^{-j\theta_r}e^{-j\beta z})\right]\right\}^{1/2} \\ &= |V_0^+|\left[1 + |\Gamma|^2 + |\Gamma|(e^{j(2\beta z + \theta_r)} + e^{-j(2\beta z + \theta_r)})\right]^{1/2} \\ &= |V_0^+|\left[1 + |\Gamma|^2 + 2|\Gamma|\cos(2\beta z + \theta_r)\right]^{1/2}, \end{aligned} \tag{2.64}$$

where we have used the identity

$$e^{jx} + e^{-jx} = 2\cos x \tag{2.65}$$

for any real quantity $x$. To express the magnitude of $\widetilde{V}$ as a function of $d$ instead of $z$, we replace $z$ with $-d$ on the right-hand side of Eq. (2.64):

$$|\widetilde{V}(d)| = |V_0^+|\left[1 + |\Gamma|^2 + 2|\Gamma|\cos(2\beta d - \theta_r)\right]^{1/2}. \tag{2.66}$$

By applying the same steps to Eq. (2.63b), a similar expression can be derived for $|\widetilde{I}(d)|$, which is the magnitude of the current $\widetilde{I}(d)$:

$$|\widetilde{I}(d)| = \frac{|V_0^+|}{Z_0}\left[1 + |\Gamma|^2 - 2|\Gamma|\cos(2\beta d - \theta_r)\right]^{1/2}. \tag{2.67}$$

The variations of $|\widetilde{V}(d)|$ and $|\widetilde{I}(d)|$ as a function of $d$, which is the position on the line relative to the load (at $d = 0$), are illustrated in **Fig. 2-14** for a line with $|V_0^+| = 1$ V, $|\Gamma| = 0.3$, $\theta_r = 30^\circ$, and $Z_0 = 50$ Ω. The sinusoidal patterns are called ***standing waves*** and are caused by the *interference* of the two traveling waves. The *maximum value* of the ***standing-wave pattern*** of $|\widetilde{V}(d)|$ corresponds to the position on the line at which the incident and reflected waves are ***in-phase*** [$2\beta d - \theta_r = 2n\pi$ in Eq. (2.66)]. Therefore, they add constructively to give a value equal to $(1 + |\Gamma|)|V_0^+| = 1.3$ V. The *minimum value* of $|\widetilde{V}(d)|$ occurs when the two waves interfere destructively, which occurs when the incident and reflected waves are in ***phase-opposition*** [$2\beta d - \theta_r = (2n+1)\pi$]. In this case, $|\widetilde{V}(d)| = (1 - |\Gamma|)|V_0^+| = 0.7$ V.

► Whereas the repetition period is $\lambda$ for the incident and reflected waves considered individually, the repetition period of the standing-wave pattern is $\lambda/2$. ◄



示例 2-4：纯无功负载的 |Γ|

证明连接到纯无功负载的无损线路的 $|\Gamma| = 1$。

解：纯无功负载的负载阻抗为$Z_L = jX_L$。从方程。 (2.59)，反射系数为

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{jX_L - Z_0}{jX_L + Z_0} = \frac{-(Z_0 - jX_L)}{(Z_0 + jX_L)} = \frac{-\sqrt{Z_0^2 + X_L^2}\, e^{-j\theta}}{\sqrt{Z_0^2 + X_L^2}\, e^{j\theta}} = -e^{-j2\theta} = 1e^{j(\pi - 2\theta)},$$

其中 $\theta = \tan^{-1} X_L/Z_0$。因此，对于 $\Gamma = |\Gamma|e^{j\theta_r}$，

$$|\Gamma| = 1$$

和

$$\theta_r = \pi - 2\theta = \pi - 2\tan^{-1}\left(\frac{X_L}{Z_0}\right).$$

练习 2-8：50 Ω 无损传输线的端接阻抗为 $Z_L = (30 - j200)$ Ω 的负载。计算负载处的电压反射系数。

**Answer:** $\Gamma = 0.93/-27.5^\circ$. (See EM.)

练习 2-9：150 Ω 无损线路的端接阻抗为 $Z_L = -j30$ Ω 的电容器。计算Γ。

**Answer:** $\Gamma = 1/-157.4^\circ$. (See EM.)

练习 2-10：假设负载处的反射系数为 $\Gamma = 0.6 - j0.3$，请确定归一化负载阻抗 $z_L$。

**Answer:** $z_L = 2.2 - j2.4$. (See EM.)

## 2-6.2 驻波

使用等式中的关系 $V_0^- = \Gamma V_0^+$ (2.54a) 和 (2.54b) 产量

$$\widetilde{V}(z) = V_0^+(e^{-j\beta z} + \Gamma e^{j\beta z}), \tag{2.63a}$$

$$\widetilde{I}(z) = \frac{V_0^+}{Z_0}(e^{-j\beta z} - \Gamma e^{j\beta z}). \tag{2.63b}$$

这些表达式现在仅包含一个（尚未确定）未知数 $V_0^+$。然而，在我们继续求解 $V_0^+$ 之前，让我们检查一下这些表达式背后的物理意义。我们首先推导 $|\widetilde{V}(z)|$ 的表达式，它是 $\widetilde{V}(z)$ 的大小。使用方程式后。 (2.62) 式(2.63a) 并应用关系式 $|\widetilde{V}(z)| = [\widetilde{V}(z)\widetilde{V}^*(z)]^{1/2}$，其中 $\widetilde{V}^*(z)$ 是 $\widetilde{V}(z)$ 的复共轭，我们有

$$\begin{aligned}|\widetilde{V}(z)| &= \left\{\left[V_0^+(e^{-j\beta z} + |\Gamma|e^{j\theta_r}e^{j\beta z})\right] \cdot \left[(V_0^+)^*(e^{j\beta z} + |\Gamma|e^{-j\theta_r}e^{-j\beta z})\right]\right\}^{1/2} \\ &= |V_0^+|\left[1 + |\Gamma|^2 + |\Gamma|(e^{j(2\beta z + \theta_r)} + e^{-j(2\beta z + \theta_r)})\right]^{1/2} \\ &= |V_0^+|\left[1 + |\Gamma|^2 + 2|\Gamma|\cos(2\beta z + \theta_r)\right]^{1/2}, \end{aligned} \tag{2.64}$$

我们在哪里使用了身份

$$e^{jx} + e^{-jx} = 2\cos x \tag{2.65}$$

对于任何实际数量 x。为了将 $\widetilde{V}$ 的大小表示为 d 而不是 z 的函数，我们将等式右侧的 z 替换为 −d。 (2.64)：

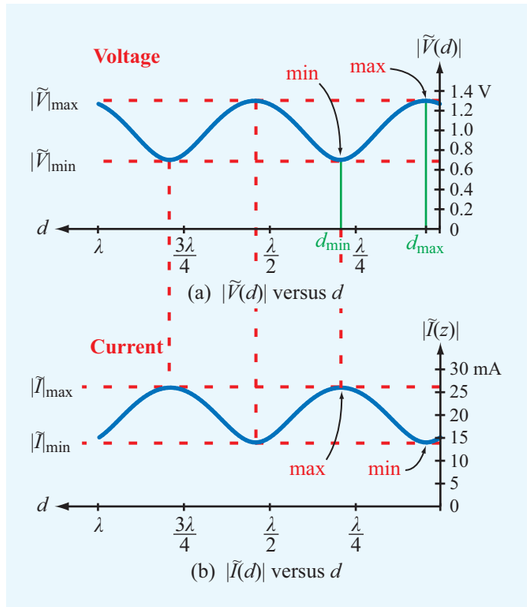
$$|\widetilde{V}(d)| = |V_0^+|\left[1 + |\Gamma|^2 + 2|\Gamma|\cos(2\beta d - \theta_r)\right]^{1/2}. \tag{2.66}$$

通过对等式应用相同的步骤。 (2.63b)，对于$|\widetilde{I}(d)|$可以导出类似的表达式，它是电流I(d)的幅度：

$$|\widetilde{I}(d)| = \frac{|V_0^+|}{Z_0}\left[1 + |\Gamma|^2 - 2|\Gamma|\cos(2\beta d - \theta_r)\right]^{1/2}. \tag{2.67}$$

$|\widetilde{V}(d)|$ 和 $|\widetilde{I}(d)|$ 的变化作为 d 的函数，d 是线上相对于负载的位置（在 d = 0 处），如图 2-14 所示，其中一条线为 $|V_0^+| = 1$ V、$|\Gamma| = 0.3$、$\theta_r = 30^\circ$ 和 $Z_0 = 50$ Ω。正弦曲线称为驻波，是由两个行波干涉引起的。 $|\widetilde{V}(d)|$ 驻波图的最大值对应于等式中入射波和反射波同相的线上位置 [$2\beta d - \theta_r = 2n\pi$。 (2.66)]。因此，它们相加后得到的值等于 $(1 + |\Gamma|)|V_0^+| = 1.3$ V。当两个波相消干涉时，即当入射波和反射波相位相反时，会出现 $|\widetilde{V}(d)|$ 的最小值 [$2\beta d - \theta_r = (2n + 1)\pi$]。在这种情况下，$|\widetilde{V}(d)| = (1 - |\Gamma|)|V_0^+| = 0.7$ V。

► 单独考虑入射波和反射波的重复周期为 λ，而驻波图案的重复周期为 λ/2。 ◄



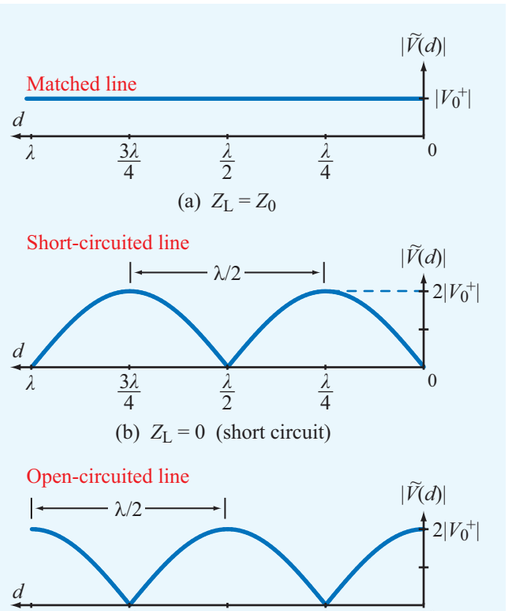




**Figure 2-14** Standing-wave pattern for (a) $|\tilde{V}(d)|$ and (b) $|\tilde{I}(d)|$ for a lossless transmission line of characteristic impedance $Z_0 = 50\ \Omega$, terminated in a load with a reflection coefficient $\Gamma = 0.3e^{j30^\circ}$. The magnitude of the incident wave $|V_0^+| = 1$ V. The standing-wave ratio is $S = |\tilde{V}|_{max}/|\tilde{V}|_{min} = 1.3/0.7 = 1.86$.

**Figure 2-15** Voltage standing-wave patterns for (a) a matched load, (b) a short-circuited line, and (c) an open-circuited line.

The standing-wave pattern describes the spatial variation of the magnitude of $\tilde{V}(d)$ as a function of $d$. If one were to observe the variation of the instantaneous voltage as a function of time at location $d = d_{max}$ in **Fig. 2-14**, that variation would be as $\cos\omega t$ and would have an amplitude equal to 1.3 V [i.e., $v(t)$ would oscillate between $-1.3$ V and $+1.3$ V]. Similarly, the instantaneous voltage $v(d,t)$ at any location $d$ will be sinusoidal with amplitude equal to $|\tilde{V}(d)|$ at that $d$.

► Interactive **Module 2.4**[a] provides a highly recommended simulation tool for gaining better understanding of the standing-wave patterns for $\tilde{V}(d)$ and $\tilde{I}(d)$ and the dynamic behavior of $v(d,t)$ and $i(d,t)$. ◄

[a] At em8e.eecs.umich.edu

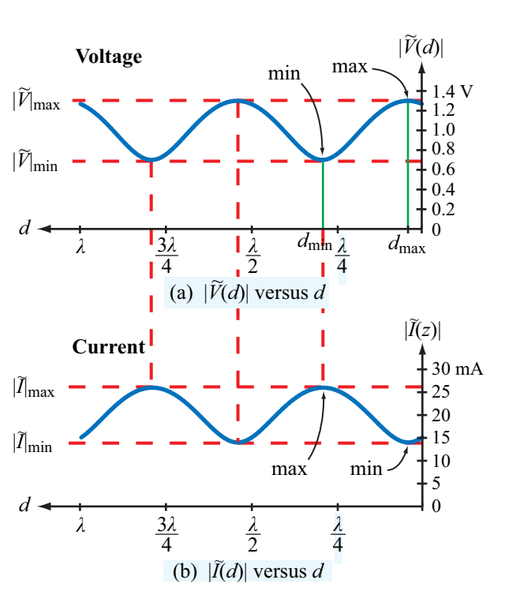
Close inspection of the voltage and current standing-wave patterns shown in **Fig. 2-14** reveals that the two patterns are in phase opposition (when one is at a maximum, the other is at a minimum, and vice versa). This is a consequence of the fact that the third term in Eq. (2.66) is preceded by a plus sign, whereas the third term in Eq. (2.67) is preceded by a minus sign.

The standing-wave patterns shown in **Fig. 2-14** are for a line with $\Gamma = 0.3e^{j30^\circ}$. The peak-to-peak variation of the pattern ($|\tilde{V}|_{min} = (1 - |\Gamma|)|V_0^+|$ to $|\tilde{V}|_{max} = (1 + |\Gamma|)|V_0^+|$) depends on $|\Gamma|$. For the special case of a matched line with $Z_L = Z_0$, we have $|\Gamma| = 0$ and $|\tilde{V}(d)| = |V_0^+|$ for all values of $d$, as shown in **Fig. 2-15(a)**.

► With no reflected wave present, there is no interference and no standing waves. ◄







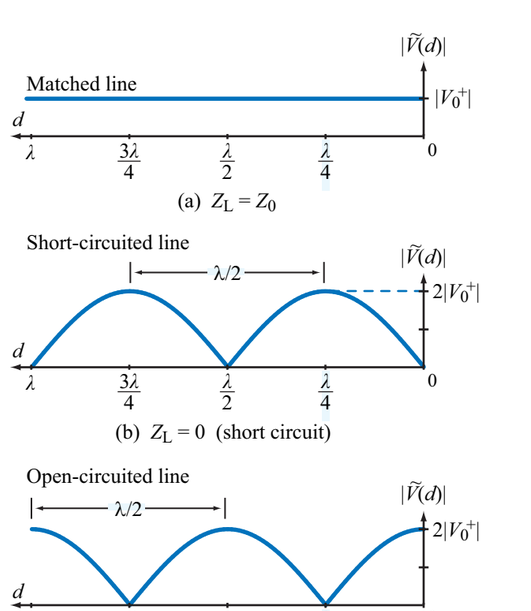




图 2-14 特性阻抗 $Z_0 = 50\ \Omega$ 的无损传输线的 (a) $|\tilde{V}(d)|$ 和 (b) $|\tilde{I}(d)|$ 的驻波图，端接于反射系数 $\Gamma = 0.3e^{j30^\circ}$ 的负载。入射波的幅度为 $|V_0^+| = 1$ V。驻波比为 $S = |\tilde{V}|_{max}/|\tilde{V}|_{min} = 1.3/0.7 = 1.86$。

图 2-15 (a) 匹配负载、(b) 短路线路和 (c) 开路线路的电压驻波模式。

驻波图将 $\tilde{V}(d)$ 幅度的空间变化描述为 d 的函数。如果观察图 2-14 中位置 $d = d_{max}$ 处的瞬时电压随时间的变化，则该变化将是 $\cos\omega t$，并且幅度等于 1.3 V [即，$v(t)$ 将在 $-1.3$ V 和 $+1.3$ V 之间振荡]。类似地，任何位置 d 处的瞬时电压 $v(d,t)$ 将是正弦波，其幅度等于该 d 处的 $|\tilde{V}(d)|$。

► 交互模块 2.4[a] 提供了强烈推荐的模拟工具，可以更好地理解 $\tilde{V}(d)$ 和 $\tilde{I}(d)$ 的驻波模式以及 $v(d,t)$ 和 $i(d,t)$ 的动态行为。 ◄

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仔细检查图 2-14 所示的电压和电流驻波模式可以发现，这两种模式是

反相（当一个处于最大值时，另一个处于最小值，反之亦然）。这是方程中第三项的结果。 (2.66) 前面有一个加号，而等式中的第三项。 (2.67) 前面有一个减号。

图 2-14 所示的驻波图是针对 $\Gamma = 0.3\,e^{j30^\circ}$ 的线。模式的峰到峰变化（$|\tilde{V}|_{min} = (1 - |\Gamma|)|V_0^+|$ 到 $|\tilde{V}|_{max} = (1 + |\Gamma|)|V_0^+|$）取决于 $|\Gamma|$。对于具有 $Z_L = Z_0$ 的匹配线的特殊情况，对于所有 d 值，我们有 $|\Gamma| = 0$ 和 $|\tilde{V}(d)| = |V_0^+|$，如图 2-15(a) 所示。

► 由于不存在反射波，因此不存在干扰，也不存在驻波。 ◄

The other end of the $|\Gamma|$ scale, at $|\Gamma| = 1$, corresponds to when the load is a short circuit ($\Gamma = -1$) or an open circuit ($\Gamma = 1$). The standing-wave patterns for those two cases are shown in **Figs. 2-15(b)** and **(c)**; both exhibit maxima of $2|V_0^+|$ and minima equal to zero, but the two patterns are spatially shifted relative to each other by a distance of $\lambda/4$. A purely reactive load (capacitor or inductor) also satisfies the condition $|\Gamma| = 1$, but $\theta_r$ is generally neither zero nor 180° (**Table 2-3**). Exercise 2.9 examines the standing-wave pattern for a lossless line terminated in an inductor.

Now let us examine the maximum and minimum values of the voltage magnitude. From Eq. (2.66), $|\widetilde{V}(d)|$ is a maximum when the argument of the cosine function is equal to zero or a multiple of $2\pi$. Let us *denote $d_{\max}$ as the distance from the load at which $|\widetilde{V}(d)|$ is a maximum*. It then follows that

$$|\widetilde{V}(d)| = |\widetilde{V}|_{\max} = |V_0^+|[1 + |\Gamma|], \tag{2.68}$$

when

$$2\beta d_{\max} - \theta_r = 2n\pi, \tag{2.69}$$

with $n = 0$ or a positive integer. Solving Eq. (2.69) for $d_{\max}$, we have

$$d_{\max} = \frac{\theta_r + 2n\pi}{2\beta} = \frac{\theta_r \lambda}{4\pi} + \frac{n\lambda}{2}, \quad \begin{cases} n = 1, 2, \ldots & \text{if } \theta_r < 0, \\ n = 0, 1, 2, \ldots & \text{if } \theta_r \geq 0, \end{cases} \tag{2.70}$$

where we have used $\beta = 2\pi/\lambda$. The phase angle of the voltage reflection coefficient, $\theta_r$, is bounded between $-\pi$ and $\pi$ radians. If $\theta_r \geq 0$, the ***first voltage maximum*** occurs at $d_{\max} = \theta_r \lambda/4\pi$, corresponding to $n = 0$, but if $\theta_r < 0$, the first physically meaningful maximum occurs at $d_{\max} = (\theta_r \lambda/4\pi) + \lambda/2$, corresponding to $n = 1$. Negative values of $d_{\max}$ correspond to locations past the end of the line; therefore, they have no physical significance.

Similarly, the minima of $|\widetilde{V}(d)|$ occur at distances $d_{\min}$ where the argument of the cosine function in Eq. (2.66) is equal to $(2n + 1)\pi$, which gives the result

$$|\widetilde{V}|_{\min} = |V_0^+|[1 - |\Gamma|], \quad \text{when } (2\beta d_{\min} - \theta_r) = (2n + 1)\pi, \tag{2.71}$$

with $-\pi \leq \theta_r \leq \pi$. The first minimum corresponds to $n = 0$. The spacing between a maximum $d_{\max}$ and the adjacent minimum $d_{\min}$ is $\lambda/4$. Hence, the ***first minimum*** occurs at

$$d_{\min} = \begin{cases} d_{\max} + \lambda/4, & \text{if } d_{\max} < \lambda/4, \\ d_{\max} - \lambda/4, & \text{if } d_{\max} \geq \lambda/4. \end{cases} \tag{2.72}$$

▶ The locations on the line corresponding to voltage maxima correspond to current minima, and vice versa. ◀

The ratio of $|\widetilde{V}|_{\max}$ to $|\widetilde{V}|_{\min}$ is called the ***voltage standing-wave ratio*** $S$, which from Eqs. (2.68) and (2.71) is given by

$$S = \frac{|\widetilde{V}|_{\max}}{|\widetilde{V}|_{\min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|} \quad \textbf{(dimensionless)}. \tag{2.73}$$

This quantity is often referred to by its acronym, VSWR or the shorter acronym ***SWR***, and it provides a measure of the mismatch between the load and the transmission line:

▶ For a matched load with $\Gamma = 0$, we get $S = 1$, and for a line with $|\Gamma| = 1$, $S = \infty$. ◀

**Concept Question 2-5:** The attenuation constant $\alpha$ represents ohmic losses. In view of the model given in **Fig. 2-6(c)**, what should $R'$ and $G'$ be in order to have no losses? Verify your expectation through the expression for $\alpha$ given by Eq. (2.25a).

**Concept Question 2-6:** How is the wavelength $\lambda$ of the wave traveling on the transmission line related to the free-space wavelength $\lambda_0$?

**Concept Question 2-7:** When is a load matched to a transmission line? Why is it important?

**Concept Question 2-8:** What is a standing-wave pattern? Why is its period $\lambda/2$ and not $\lambda$?

**Concept Question 2-9:** What is the separation between the location of a voltage maximum and the adjacent current maximum on the line?





$|\Gamma|$ 标度的另一端 $|\Gamma| = 1$ 处对应于负载短路 ($\Gamma = -1$) 或开路 ($\Gamma = 1$) 的情况。这两种情况的驻波模式如图 2 和 3 所示。 2-15(b)和(c)；两者都表现出最大值 $2|V_0^+|$ 和最小值等于 0，但是这两种模式在空间上相对于彼此偏移了距离 $\lambda/4$。纯无功负载（电容器或电感器）也满足条件 $|\Gamma| = 1$，但 $\theta_r$ 通常既不为零也不 180° (表 2-3)。练习 2.9 检查终止于电感器的无损线路的驻波模式。

现在让我们检查电压幅度的最大值和最小值。从方程。 (2.66)，当余弦函数的自变量等于0或$2\pi$的倍数时，$|\widetilde{V}(d)|$是最大值。让我们将 $d_{\max}$ 表示为与 $|\widetilde{V}(d)|$ 达到最大值时的负载的距离。那么接下来

$$|\widetilde{V}(d)| = |\widetilde{V}|_{\max} = |V_0^+|[1 + |\Gamma|], \tag{2.68}$$

什么时候

$$2\beta d_{\max} - \theta_r = 2n\pi, \tag{2.69}$$

其中 n = 为 0 或正整数。求解方程。 (2.69) 对于 $d_{\max}$，我们有

$$d_{\max} = \frac{\theta_r + 2n\pi}{2\beta} = \frac{\theta_r \lambda}{4\pi} + \frac{n\lambda}{2}, \quad \begin{cases} n = 1, 2, \ldots & \text{if } \theta_r < 0, \\ n = 0, 1, 2, \ldots & \text{if } \theta_r \geq 0, \end{cases} \tag{2.70}$$

我们使用了 $\beta = 2\pi/\lambda$。电压反射系数的相角 $\theta_r$ 介于 $-\pi$ 和 $\pi$ 弧度之间。如果 $\theta_r \geq 0$，则第一个电压最大值出现在 $d_{\max} = \theta_r \lambda/4\pi$ 处，对应于 $n = 0$，但如果 $\theta_r < 0$，则第一个有物理意义的最大值出现在 $d_{\max} = (\theta_r \lambda/4\pi) + \lambda/2$ 处，对应于 $n = 1$。$d_{\max}$ 的负值对应于超过行尾的位置；因此，它们没有物理意义。

类似地，$|\widetilde{V}(d)|$ 的最小值出现在距离 $d_{\min}$ 处，其中等式中余弦函数的参数如下： (2.66) 等于 $(2n + 1)\pi$，得出结果

$$|\widetilde{V}|_{\min} = |V_0^+|[1 - |\Gamma|], \quad \text{when } (2\beta d_{\min} - \theta_r) = (2n + 1)\pi, \tag{2.71}$$

与$-\pi \leq \theta_r \leq \pi$。第一个最小值对应于 $n = 0$。最大值 $d_{\max}$ 与相邻最小值 $d_{\min}$ 之间的间距为 $\lambda/4$。因此，第一个最小值出现在

$$d_{\min} = \begin{cases} d_{\max} + \lambda/4, & \text{if } d_{\max} < \lambda/4, \\ d_{\max} - \lambda/4, & \text{if } d_{\max} \geq \lambda/4. \end{cases} \tag{2.72}$$

▶ 线上对应于电压最大值的位置对应于电流最小值，反之亦然。 ◀

$|\widetilde{V}|_{\max}$ 与 $|\widetilde{V}|_{\min}$ 之比称为电压驻波比 S，由等式 1 得出。 (2.68) 和 (2.71) 由下式给出

$$S = \frac{|\widetilde{V}|_{\max}}{|\widetilde{V}|_{\min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|} \quad \textbf{(dimensionless)}. \tag{2.73}$$

该量通常用其缩写词 VSWR 或更短的缩写词 SWR 来表示，它提供了负载与传输线之间不匹配的度量：

▶ 对于 $\Gamma = 0$ 的匹配负载，我们得到 S = 1，对于 $|\Gamma| = 1$ 的线，我们得到 S = ∞。 ◀

概念问题 2-5：衰减常数 α 代表欧姆损耗。鉴于图2-6(c)给出的模型，为了没有损失，R′和G′应该是多少？通过式（1）给出的 α 表达式验证您的期望。 （2.25a）。

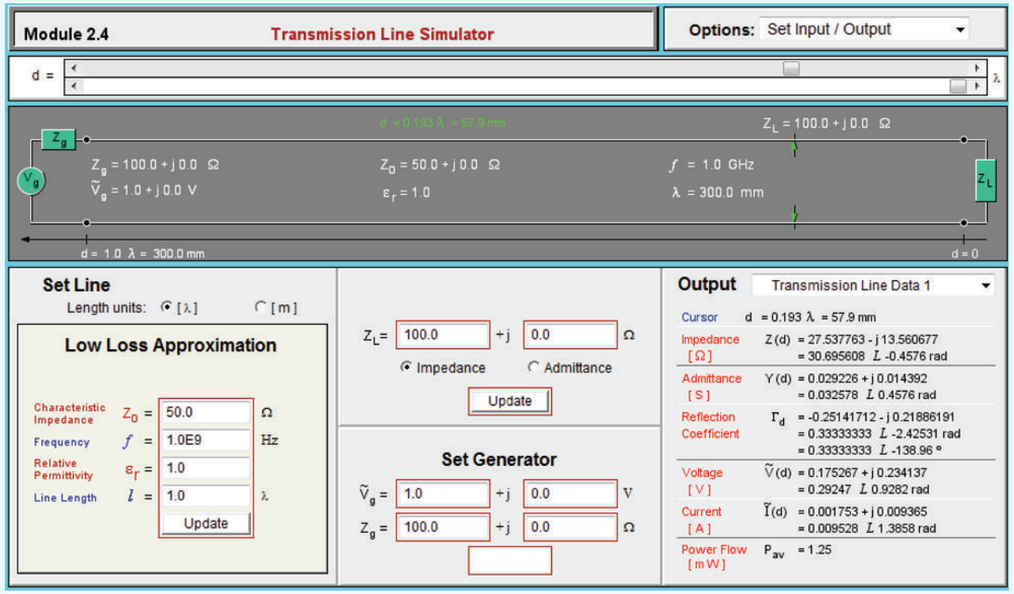
概念问题 2-6：在传输线上传播的波的波长 λ 与自由空间波长 $\lambda_0$ 有何关系？

概念问题 2-7：负载何时与传输线匹配？为什么它很重要？

概念问题 2-8：什么是驻波模式？为什么它的周期是 $\lambda/2$而不是λ？

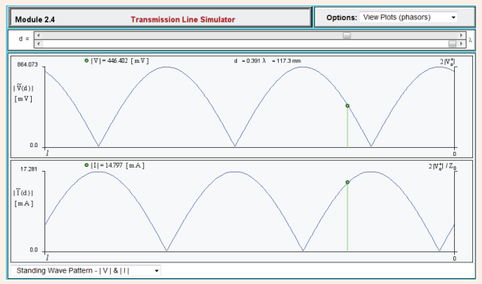
概念问题 2-9：线路上电压最大值的位置与相邻电流最大值的位置之间的间隔是多少？

**Module 2.4 Transmission Line Simulator** Upon specifying the requisite input data—including the load impedance at $d = 0$ and the generator voltage and impedance at $d = l$, this module provides a wealth of output information about the voltage and current waveforms along the transmission line. You can view plots of the standing-wave patterns for voltage and current, the time and spatial variations of the instantaneous voltage $\upsilon(d,t)$ and current $i(d,t)$, and other related quantities.




**Exercise 2-11:** Use Module 2.4 to generate the voltage and current standing-wave patterns for a 50 Ω line of length 1.5λ terminated in an inductance with $Z_L = j140\ \Omega$.

**Answer:** See Module 2.4 display.




**Example 2-5: Standing-Wave Ratio**

A 50 Ω transmission line is terminated in a load with $Z_L = (100 + j50)\ \Omega$. Find the voltage reflection coefficient and the voltage standing-wave ratio.

**Solution:** From Eq. (2.59), Γ is given by

$$\Gamma = \frac{z_L - 1}{z_L + 1} = \frac{(2 + j1) - 1}{(2 + j1) + 1} = \frac{1 + j1}{3 + j1}.$$

Converting the numerator and denominator to polar form yields

$$\Gamma = \frac{1.414e^{j45^\circ}}{3.162e^{j18.4^\circ}} = 0.45e^{j26.6^\circ}.$$

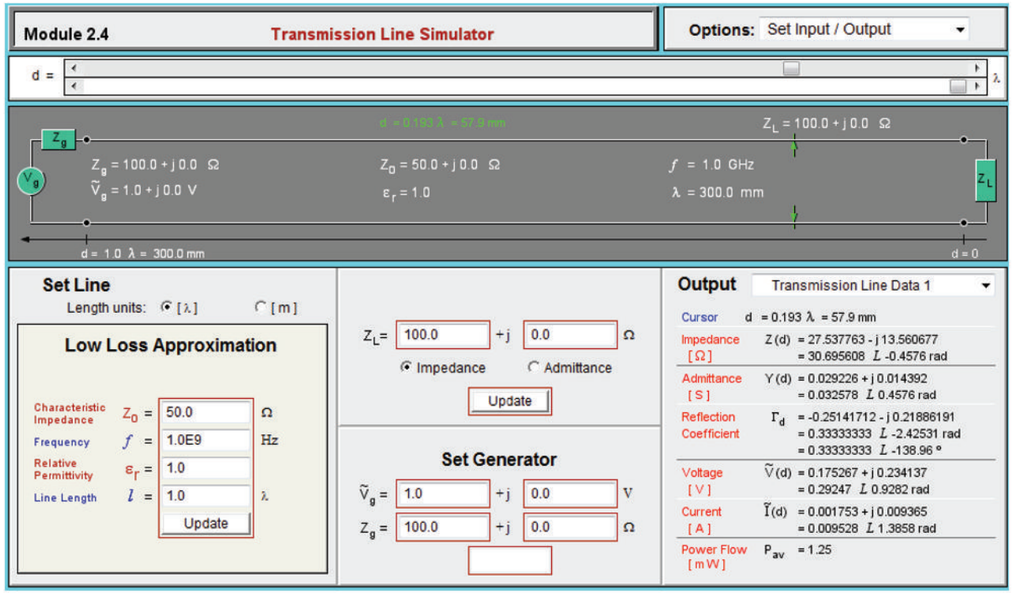
Using the definition for $S$ given by Eq. (2.73), we have

$$S = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + 0.45}{1 - 0.45} = 2.6.$$



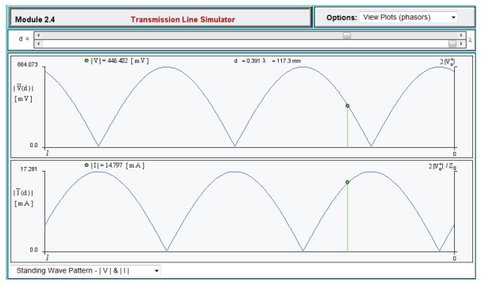
70 CHAPTER 2 TRANSMISSION LINES

模块 2.4 传输线模拟器 在指定必要的输入数据（包括 d = 0 处的负载阻抗以及 d = l 处的发电机电压和阻抗）后，该模块提供了有关沿传输线的电压和电流波形的丰富输出信息。您可以查看电压和电流的驻波模式图、瞬时电压 υ(d,t) 和电流 i(d,t) 的时间和空间变化以及其他相关量。




练习 2-11：使用模块 2.4 生成长度为 1.5λ 的 50 Ω 线的电压和电流驻波模式，该线终止于 $Z_L = j140\ \Omega$ 的电感。

答：参见模块2.4显示。




例 2-5：驻波比

50 Ω 传输线的端接负载为 $Z_L = (100 + j50)\ \Omega$。求电压反射系数和电压驻波比。

解决方案：从方程。 (2.59), Γ 由下式给出

$$\Gamma = \frac{z_L - 1}{z_L + 1} = \frac{(2 + j1) - 1}{(2 + j1) + 1} = \frac{1 + j1}{3 + j1}.$$

将分子和分母转换为极坐标形式得到

$$\Gamma = \frac{1.414e^{j45^\circ}}{3.162e^{j18.4^\circ}} = 0.45e^{j26.6^\circ}.$$

使用方程式给出的 S 定义： (2.73) ，我们有

$$S = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + 0.45}{1 - 0.45} = 2.6.$$